

Luma – Hardware Simulator Realtime Database Schema

SCS 3311 – Mobile Application Design and Development

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1 Overview

Luma's backend is a single Firebase Realtime Database organized as one JSON tree. The top-level nodes used by the Hardware Simulator are **users**, **devices**, **alerts**, **reports**, and **userLogs**. Floors and rooms are nested under **users/{uid}**, while device *records* live flat under **devices** and are referenced by ID from within the floor/room structure. This deliberate denormalization follows Firebase's recommended practice of flattening deeply nested data so that listening to a single device downloads only that device's record rather than an entire room or floor subtree.

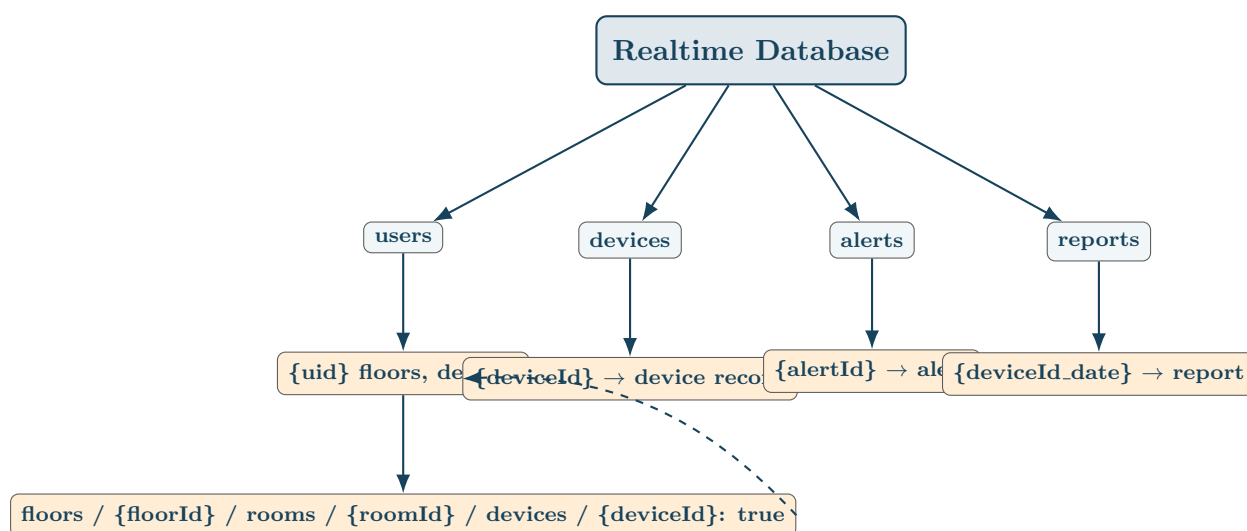


Figure 1: Top-level database structure. The dashed arrow indicates a device-ID reference rather than nested ownership.

2 The devices Node

Each simulated device is stored at **devices/{deviceId}**. A device ID is auto-generated by the Simulator using the pattern `device-<timestamp>-<random>` and a new record is created by `addDevice()` in `deviceService.ts`. The record is written into the database with the following structure, where `{...typeFields}` varies by device type:

```

1 "devices": {
2   "device001": {
3     "type": "<type>",
4     "name": "Living Room Light",
5     "state": "ON | OFF",
6     "status": "ON | OFF | ONLINE | ERROR | DISCONNECTED",
7     "lastSeen": 1739000000000,
8     "userId": "UID",
  
```

```

9     "...typeFields": { }
10   }
11 }

```

Listing 1: Generic shape of a device record.

2.1 Common Fields

Table 1 lists the fields shared by all five device types (`outlet`, `light`, `iron`, `switchPanel`, `camera`).

Table 1: Fields common to every device record.

Field	Type	Values / Units	Description
<code>type</code>	string	<code>outlet</code> , <code>light</code> , <code>iron</code> , <code>switchPanel</code> , <code>camera</code>	Discriminator that selects the device-specific fields and controls.
<code>name</code>	string	free text	Human-readable device name shown in both clients.
<code>state</code>	string	ON / OFF	Power state written by the on/off control. Absent for cameras (which are always “streaming”).
<code>status</code>	string	ON / OFF / ONLINE / ERROR / DISCONNECTED	Connectivity/health status; cameras default to ONLINE.
<code>lastSeen</code>	number	epoch ms	Heartbeat timestamp; written on every state change. Consumed by the disconnection-detector Cloud Function.
<code>userId</code>	string	Firebase UID	Owning user; used to scope the per-user device index.

2.2 Type-Specific Fields

The tables below list the extra fields written for each device type, as produced by `AddDeviceDialog.tsx` and consumed by the Cloud Functions.

2.2.1 Electrical Outlet (`outlet`)

An outlet has no extra fields beyond the common ones; it is the baseline ON/OFF device.

2.2.2 Smart Light (`light`)

Table 2: Fields specific to `light` devices.

Field	Type	Values / Units	Description
<code>startTime</code>	string	"HH:MM"	Automatic on time, consumed by the light-scheduler Cloud Function.
<code>endTime</code>	string	"HH:MM"	Automatic off time, consumed by the light-scheduler Cloud Function.
<code>turnedOnAt</code>	number — null	epoch ms	Set to <code>Date.now()</code> when turned on; cleared on off/shutdown.

2.2.3 Electric Iron (`iron`)

Table 3: Fields specific to `iron` devices.

Field	Type	Values / Units	Description
<code>turnedOnAt</code>	number — null	epoch ms	Time the iron was switched on; used to measure elapsed run time.
<code>maxDurationMinutes</code>	number	5–120 (step 5, default 30)	Safety auto-shutdown limit enforced by the iron auto-shutdown Cloud Function.

2.2.4 Multi-Switch Panel (`switchPanel`)

Table 4: Fields specific to `switchPanel` devices.

Field	Type	Values / Units	Description
<code>switchCount</code>	number	2–5 (default 3)	Number of physical switches on the panel.
<code>switches</code>	object	{ <code>switch1</code> : "ON OFF", ...}	One key per switch (<code>switch1...switchN</code>); each toggled independently via <code>setSwitchState()</code> .

2.2.5 Camera (camera)

Table 5: Fields specific to **camera** devices.

Field	Type	Values / Units	Description
<code>lastSnapshotUrl</code>	string	HTTPS URL	Most recent mock image URL (generated via <code>picsum.photos</code>).
<code>lastSnapshotAt</code>	number	epoch ms	Time the snapshot was requested.

2.2.6 Power Usage History (history)

Written by the report-aggregator Cloud Function when a device transitions **ON** → **OFF**. Each entry is pushed onto `devices/{deviceId}/history`:

Table 6: Fields of each **history** entry.

Field	Type	Values / Units	Description
<code>turnedOffAt</code>	number	epoch ms	When the device was switched off.
<code>durationMinutes</code>	number	minutes	Length of the on-session, rounded to the nearest minute.
<code>estimatedKwh</code>	number	kWh (3 d.p.)	Estimated energy consumed, computed from nominal wattage × duration.

3 The users Node

Floors, rooms, and a per-user device index live under `users/{uid}`. Room membership is expressed as a set of device-ID references (`{deviceId}: true`); the device records themselves remain in the flat **devices** node.

Table 7: Structure of the **users** node.

Path	Type	Value	Description
users/{uid}/floors/{floorId}/name	string	free text	Floor display name (e.g. “Ground Floor”).
users/{uid}/floors/{floorId}/icon	string	home, stairs, attic, ...	Icon key derived by the clients.
users/{uid}/floors/{floorId}/rooms/{roomId}/name	string	free text	Room display name (e.g. “Living Room”).
users/{uid}/floors/{floorId}/rooms/{roomId}/devices/{deviceId}	boolean	true	Membership reference to a device record.
users/{uid}/devices/{deviceId}	boolean	true	Flattened index of all devices owned by the user.

4 The alerts Node

Alerts are pushed by the Cloud Functions (iron auto-shutdown, disconnection detector) and read by the Simulator’s alert list. Each record is keyed by an auto-generated push ID.

Table 8: Fields of an `alerts` record.

Field	Type	Values / Units	Description
<code>deviceId</code>	string	device ID	The device the alert concerns.
<code>type</code>	string	AUTO_SHUTDOWN, DEVICE_DISCONNECTED	Machine-readable alert category.
<code>message</code>	string	free text	Human-readable alert description.
<code>timestamp</code>	number	epoch ms	When the alert was raised.

5 The reports Node

Usage reports are written exclusively by the report-aggregator Cloud Function (client writes are denied by the security rules). Each record is keyed `{deviceId}_{YYYY-MM-DD}` and aggregates a device's sessions for the day, week, and month.

Table 9: Fields of a `reports` record.

Field	Type	Values / Units	Description
<code>deviceId</code>	string	device ID	The device this report covers.
<code>deviceName</code>	string	free text	Device name snapshot at aggregation time.
<code>deviceType</code>	string	one of the five types	Device type snapshot.
<code>period</code>	string	daily	Reporting period of the record.
<code>date</code>	string	YYYY-MM-DD	Day the record covers.
<code>week</code>	string	YYYY-MM-DD (week start)	Week the record contributes to.
<code>month</code>	string	YYYY-MM	Month the record contributes to.
<code>totalOnDurationMinutes</code>	number	minutes	Cumulative on-duration for the day.
<code>totalSessions</code>	number	count	Number of on/off sessions recorded.
<code>estimatedKwh</code>	number	kWh (3 d.p.)	Cumulative estimated energy usage.
<code>lastUpdated</code>	number	epoch ms	Last time the report was recomputed.

6 The userLogs Node

The Simulator records a usage/audit log for each signed-in user. Every entry is pushed onto `userLogs/{uid}` by the `logEvent()` callback in `App.tsx`.

Table 10: Fields of a userLogs entry.

Field	Type	Values / Units	Description
environment	string	simulator	Client that produced the log entry.
device	string	user agent string	Browser/platform of the client.
event	string	e.g. <code>session_start</code> , <code>device_toggle</code>	Event name.
sessionId	string	random UUID	Identifies a browser session.
timestamp	number	epoch ms	When the event occurred.
details	object	key/value metadata	Per-event context (e.g. {deviceId, state}).

7 Complete JSON Example

The listing below shows the full tree written by `seedSampleData()` together with the alert, report, and log records produced by the Cloud Functions and Simulator.

```

1 {
2   "devices": {
3     "device001": {
4       "type": "light", "name": "Living Room Light",
5       "state": "OFF", "status": "OFF",
6       "startTime": "18:00", "endTime": "23:00",
7       "turnedOnAt": null,
8       "lastSeen": 1739000000000, "userId": "UID"
9     },
10    "device003": {
11      "type": "iron", "name": "Bedroom Iron",
12      "state": "OFF", "status": "OFF",
13      "turnedOnAt": null, "maxDurationMinutes": 30,
14      "lastSeen": 1739000000000, "userId": "UID",
15      "history": {
16        "-0abc123": {
17          "turnedOffAt": 1739000400000,
18          "durationMinutes": 25,
19          "estimatedKwh": 0.417
20        }
21      }
22    },
23    "device004": {
24      "type": "switchPanel", "name": "Study Switch Panel",
25      "switchCount": 3,
26      "switches": { "switch1": "OFF", "switch2": "OFF", "switch3": "OFF" },
27      "status": "OFF",
28      "lastSeen": 1739000000000, "userId": "UID"
29    },
30    "device005": {
31      "type": "camera", "name": "Garage Camera",
32      "status": "ONLINE",
33      "lastSnapshotUrl": "https://picsum.photos/seed/camera-garage/640/480",
34      "lastSnapshotAt": 1739000000000,
35      "lastSeen": 1739000000000, "userId": "UID"
36    }
37  },
38  "users": {
39    "UID": {
40      "floors": {
41        "floor001": {

```

```

42     "name": "Ground Floor", "icon": "home",
43     "rooms": {
44         "room001": {
45             "name": "Living Room",
46             "devices": { "device001": true }
47         }
48     }
49 },
50 },
51 "devices": { "device001": true, "device003": true }
52 },
53 },
54 "alerts": {
55     "-Oxyz789": {
56         "deviceId": "device003",
57         "type": "AUTO_SHUTDOWN",
58         "message": "Iron \"Bedroom Iron\" automatically turned OFF after 30
59         minutes for safety",
60         "timestamp": 1739001800000
61     }
62 },
63 "reports": {
64     "device003_2026-07-10": {
65         "deviceId": "device003", "deviceName": "Bedroom Iron",
66         "deviceType": "iron", "period": "daily",
67         "date": "2026-07-10", "week": "2026-07-06", "month": "2026-07",
68         "totalOnDurationMinutes": 48, "totalSessions": 2,
69         "estimatedKwh": 0.8, "lastUpdated": 1739001800000
70     }
71 },
72 "userLogs": {
73     "UID": {
74         "-Olog456": {
75             "environment": "simulator",
76             "device": "Mozilla/5.0 (X11; Linux x86_64) ...",
77             "event": "device_toggle",
78             "sessionId": "a1b2c3d4",
79             "timestamp": 1739001200000,
80             "details": { "deviceId": "device001", "state": "ON" }
81         }
82     }
83 }

```

Listing 2: Complete database tree for the seeded home.